

K229 PRE-STAGE Lyra Friday Substrate-Mechanism Class TRIPLE Audit (Mersenne-edge / Bergman-bulk / HYBRID)

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2026-06-06 Saturday ~10:15 EDT

K229 PRE-STAGE — Lyra Friday Substrate-Mechanism Class TRIPLE Audit

0. Purpose + audit scope

K-audit on Lyra's Friday substantive headline: lepton + EW observables sort cleanly into three substrate-mechanism classes:

- substrate-Mersenne family — edge K-types, generations 0 and 3
- substrate-Bergman family — bulk K-types, generations 1 and 2
- HYBRID class — central vacuum + Higgs

Per CI_BOARD K229 Saturday priority + per Lyra morning status + per Keeper Saturday wake-board addition.

Audit scope (this is wider than a typical K-audit):

A typical K-audit examines ONE observable's substrate prediction against measurement. K229 is different — it audits an architectural meta-claim: a *partition* of multiple observables across distinct substrate-mechanism families. This is taxonomy-level, not observable-level.

The Friday six-category audit framework is observable-level. K229 candidate category SUBSTRATE-MECHANISM-FAMILY-PARTITION would be the first meta-level addition. Promotion warrants careful scrutiny.

Audit disposition tier: PRE-STAGE (not full K-audit PASS). Three substantive gates need closure before promotion to STANDING:

1. Class-definition gate: Lyra must specify (i) gen-indexing convention (0-3 generations including non-SM-generation 0+3 OR 1-indexed gen-1 + gen-2 + gen-3 as physical leptons), (ii) edge/bulk K-type formal partition (Shilov-boundary K-types vs bulk K-types per Lyra v0.5/v0.6 bulk-color work?), (iii) HYBRID class membership criterion (what makes Higgs HYBRID and not pure-Bergman or pure-Mersenne?)
2. Independence-vs-overlap gate: Mersenne tower ($M(p) = 2^p - 1$) and Bergman kernel are BOTH substrate primitives operating on the entire substrate, NOT geographically restricted. The TRIPLE classifies WHICH primitive is dominant per observable, not WHICH substrate-region. This needs explicit framing.
3. Forcing-vs-pattern-discovery gate: Does substrate FORCE 3-class partition (load-bearing structural claim), or are these 3 USE-patterns that emerge from forward derivation (descriptive observation)? K229 disposition depends on this distinction.

1. Substantive Lyra Friday content

Per Lyra Friday EOD + Lyra Saturday morning status:

1.1. Concrete forward derivations

Observable	Class	Substrate form	Precision	Substrate-tier
$m_\mu/m_e = 207$	Bergman gen-1/gen-0	Pochhammer + Bergman cascade	0.112%	Tier 2 STRUCTURAL
m_τ/m_μ	Mersenne gen-3 / Bergman gen-2	cross-family ratio	0.06%	Tier 2 STRUCTURAL
$m_H \approx 125.28 \text{ GeV}$	HYBRID central vacuum	central vacuum + Higgs forward derivation	0.14%	Tier 2 STRUCTURAL

Additional EW sector: 6 observables Tier 2 or better (Weinberg $\sin^2\theta_W = \text{rank}/N_c^2 = 2/9$ at 0.31%, $\cos^2\theta_W = g/N_c^2 = 7/9$ at 0.09%, $m_W/m_Z = \sqrt{g/N_c} = \sqrt{7/3}$ at 0.05%, $\lambda_H = 4/31$ K224, etc.).

1.2. Parallel-primitive observation (filed separately as K229b candidate)

$N_c^2 \cdot n_C = 45$ appears in BOTH: - PMNS $\sin^2\theta_{13} = 1/45$ (substrate-primary 0.46%) - CKM $\sin \theta_C \propto N_c^2/(2^N N_c \cdot n_C)$ (Lyra Friday derivation)

One substrate primitive crossing quark + lepton mixing sectors — Cal #36 STANDING positive-search Schur-generator candidate. This is filed separately as K229b because it concerns mixing-sector cross-link (parallel-primitive), not mechanism-class partition (Mersenne/Bergman/HYBRID). Distinct claim; distinct audit.

2. F1 — Mathematical class-definition correctness

2.1. Generation indexing ambiguity

Lyra writes “gens 0 and 3” + “gens 1 and 2”. The SM has 3 charged-lepton generations (e, μ , τ). Several readings possible:

Reading	Gen-0	Gen-1	Gen-2	Gen-3
(a) 0-indexed conventional	e	μ	τ	—
(b) 1-indexed conventional	—	e	μ	τ
(c) Lyra 0-indexed + 4th-gen	e	μ	τ	(non-SM 4th gen?)
(d) Lyra 0-indexed + vacuum-as-gen-0	(vacuum)	e	μ	τ
(e) Substrate K-type indexed (0,1,2,3 = K-type quantum numbers)	edge	bulk	bulk	edge

Reading (e) is closest to what Lyra likely means based on her substrate-K-type $\times$ $SU(N_c)$ tensor product theory F24 — gens 0+3 = “outer” K-types geographically near Shilov boundary, gens 1+2 = “inner” bulk K-types. But reading (e) does NOT obviously map to physical observables (e, μ , τ).

REQUIRED CLARIFICATION FROM LYRA: which indexing convention, and how does (gen-0, gen-1, gen-2, gen-3) map to (e, μ , τ , ?) physical observables?

Tentative reading for K229 PRE-STAGE: Lyra’s substrate-K-type indexing (0,1,2,3) labels K-types in K-type spectral decomposition; physical observables (e, μ , τ) reside at gens (0, 1, 2) per the “gen-0 = e edge” hypothesis. Gen-3

may be (i) m_τ as outer K-type in 4-K-type representation OR (ii) a substrate-internal 4th K-type not directly observed. Pre-stage proceeds under reading (i): gen-0 = e, gen-1 = μ , gen-2 = τ secondary K-type, gen-3 = τ primary K-type.

Under this reading: m_τ has BOTH gen-2 and gen-3 contributions (cross-family ratio m_τ/m_μ uses Mersenne gen-3 for m_τ via Mersenne tower $2^p - 1$ dominant; gen-2 contribution may be subdominant Bergman).

2.2. Edge vs bulk K-type formal partition

Lyra v0.5/v0.6 Friday bulk-color work established bulk-Shilov dual-K-type decomposition on Hardy space $H^2(D_{IV}^5)$ per Hardy decomposition. Bulk = interior holomorphic; Shilov = boundary.

Edge K-types $\cong$ K-types living on Shilov boundary $\partial_S D_{IV}^5 = S^4 \times S^1/Z_2$ (per Sunday Tier 0 v0.1.6 correction from 2D to 5D).

Bulk K-types $\cong$ K-types living on D_{IV}^5 interior (holomorphic Bergman kernel domain).

This is well-defined per Lyra's substrate ontology. K-type assignment to edge or bulk is a structural property of the K-type's substrate location, not an arbitrary classification. F1 PASSES on this gate.

Central vacuum $\cong$ K-type at "center" of D_{IV}^5 — this is the K-type minimizing the heat-semigroup eigenvalue, the substrate ground state. Higgs HYBRID lives at this center per Lyra F24 substrate-K-type $\times$ $SU(N_c)$ tensor product theory.

2.3. Class-membership criterion for individual observables

Observable	Class assignment	Rationale
m_e (gen-0)	Mersenne edge	edge K-type + Mersenne dominant
m_μ (gen-1)	Bergman bulk	bulk K-type + Bergman cascade
m_τ (gen-2, gen-3)	Mersenne edge or HYBRID	outer K-type + Mersenne 2^p ladder
m_τ/m_μ	cross-family	gen-3 Mersenne over gen-2 Bergman
m_H $\sin^2\theta_W = 2/9$	HYBRID substrate-primary	central vacuum + Higgs special rank/(rank+g) — substrate-primary, NOT classed in TRIPLE
$\alpha^{-1} = 137 + 1/28$	substrate-primary (K228)	substrate-natural correction term

Substantive observation: EW gauge observables ($\sin^2\theta_W$, m_W/m_Z , λ_H) are NOT obviously sorted into the Mersenne/Bergman/HYBRID TRIPLE. They are substrate-primary ratios from BST integers directly. The TRIPLE may be lepton-mass-specific + Higgs (HYBRID), NOT EW-sector-comprehensive.

REQUIRED CLARIFICATION FROM LYRA: does the TRIPLE cover ALL lepton + EW observables, or only lepton-mass + Higgs (with EW gauge as substrate-primary 4th class outside TRIPLE)? Honest framing matters.

Tentative PRE-STAGE framing: TRIPLE applies to lepton-mass + Higgs (the "mass-mechanism" observables). EW gauge observables (Weinberg, m_W/m_Z) are substrate-primary direct ratios, not class-partitioned within TRIPLE.

3. F2 — Independence-vs-overlap between classes

3.1. Substrate primitives are not class-restricted

Mersenne tower $M(p) = 2^p - 1$ generates: $M(\text{rank}=2) = 3 = N_c$, $M(N_c=3) = 7 = g$, $M(n_C=5) = 31$, $M(g=7) = 127$, $M(c_2=6) = 63 = 7 \cdot 9$ (composite). This tower operates on the ENTIRE substrate; not restricted to "edge K-types".

Bergman kernel $B(z, w)$ on D_{IV}^5 exists on bulk by holomorphic structure; its boundary values are Shilov-domain. Bergman is bulk-supported but its boundary behavior reaches Shilov-edge K-types via Hardy decomposition.

These primitives are not geographically restricted. So the TRIPLE classifies WHICH primitive dominates per observable, not WHICH primitive is available.

3.2. Cross-family contamination

m_τ/m_μ cross-family ratio uses BOTH Mersenne (m_τ via gen-3 Mersenne) AND Bergman (m_μ via gen-1 Bergman). So this observable belongs to BOTH classes — or rather, sits at their intersection.

This is NOT a problem for TRIPLE classification IF the framing is “observables sort by DOMINANT mechanism per class” — but it weakens the cleanness claim.

Honest framing: TRIPLE is a DOMINANT-MECHANISM classification, not a STRICT-CLASS partition. Cross-family ratios are first-order multi-class. F2 PARTIAL.

3.3. HYBRID class definition

HYBRID = “central vacuum + Higgs”. What makes the HYBRID class distinct?

- Higgs sits at central vacuum K-type of D_{IV}^5 (minimum heat-semigroup eigenvalue) — physically distinct
- Higgs forward derivation 125.28 GeV at 0.14% uses BOTH Mersenne tower (for vacuum scale) AND Bergman kernel (for holomorphic central K-type)

So HYBRID is genuinely a third mechanism class: NOT pure-Mersenne (would not get 125.28 GeV), NOT pure-Bergman (would not get vacuum-anchored mass), but a substrate-FORCED combination at central K-type.

F2 PARTIAL but HYBRID class IS genuinely distinct (not just “where pure classes fail”).

4. F3 — Substrate-mechanism forcing-vs-pattern-discovery

4.1. The strong claim: substrate FORCES 3-class partition

If TRIPLE is FORCED by substrate, this is a structural prediction: $D_{IV}^5 + 5$ BST integers FORCES exactly 3 mechanism classes (not 2, not 4, not n).

Possible forcing argument (from Lyra F24 substrate-K-type $\times$ $SU(N_c)$ tensor product theory): - Substrate-K-type spectral decomposition has FINITE number of K-types per energy band - Bulk-Shilov decomposition is BINARY (interior + boundary) - Plus “central K-type” gives THREE distinguished regions: Shilov edge + bulk interior + central vacuum - THREE substrate regions FORCE THREE mechanism classes

If this argument holds, TRIPLE is STRUCTURALLY FORCED at FRAMEWORK level.

Required for FORCING: Lyra needs to show the three regions are (i) exhaustive (no 4th region), (ii) mutually exclusive at K-type level (modulo cross-family ratios at boundaries), (iii) substrate-FORCED by D_{IV}^5 structure NOT by observable-fit a-posteriori.

4.2. The weak claim: TRIPLE is pattern-discovery

If TRIPLE is descriptive observation (we observed 3 patterns in forward derivations), it is a CANDIDATE-tier finding, not FRAMEWORK-tier. This is the honest default disposition for pre-stage.

Honest PRE-STAGE framing: K229 disposition currently sits at CANDIDATE-tier descriptive observation. Promotion to FRAMEWORK-tier requires substrate-mechanism forcing argument (Section 4.1) with Cal cold-read closure.

4.3. Cross-K-audit link to K228 + K225 substrate-Schur

K228 substrate $\alpha^{-1} = N_{\text{max}} + 1/(2 \cdot g \cdot \text{rank})$ and K225 substrate $n_s = 1 - 1/(2 \cdot g \cdot \text{rank})$ share substrate-natural composite $28 = 2 \cdot g \cdot \text{rank}$ — Cluster L candidate for Grace G15 taxonomy.

This cross-K-audit finding suggests substrate-Schur generators operate ACROSS mechanism classes (K228 EW-sector α^{-1} vs K225 inflation-sector n_s). If TRIPLE is real, substrate-Schur generators should respect the partition OR provide a structural reason for cross-class identity.

Tentative substantive interpretation: substrate-Schur $28 = 2 \cdot g \cdot \text{rank}$ may be a META-PRIMITIVE that operates above the TRIPLE classification — generating correction terms across multiple mechanism classes. If so, this REINFORCES TRIPLE at FRAMEWORK level (the meta-primitive structure supports the class partition).

5. F4 — Falsifier specification

K-audit at proper depth requires explicit falsifier.

5.1. Falsifiers that would refute the TRIPLE

1. A 4th class emerges: forward derivation of additional lepton-EW observable that fits NONE of {Mersenne-edge, Bergman-bulk, HYBRID} but requires distinct mechanism
2. Cross-family contamination so severe TRIPLE is descriptive not structural: if every observable lives at class boundaries, TRIPLE is just labeling
3. Single-primitive coverage: if Mersenne tower alone (or Bergman alone) can be shown to generate ALL observables to current precision via substrate-FORCED forward derivation, TRIPLE is OVER-FITTED
4. Bergman-bulk = HYBRID collapse: if HYBRID class can be reduced to Bergman bulk + central-K-type vacuum without distinct mechanism, TRIPLE collapses to DOUBLE (Mersenne-edge / Bergman-bulk-incl-central)
5. Substrate-Schur dominant: if K228+K225 cross-class substrate-Schur generators are shown to be the PRIMARY mechanism with Mersenne+Bergman+HYBRID as derived patterns, TRIPLE is secondary

5.2. Falsifiers within reach

Falsifier (3) is closest to testable. Lyra F24 v0.2 substrate-color² mechanism analysis multi-week per Cal #189 is exactly the work that could either (a) close a single-primitive derivation (refuting TRIPLE) or (b) confirm distinct mechanisms per class (supporting TRIPLE).

Falsifier (4) is also testable: HYBRID class is currently 1-observable (Higgs). If no other observable goes HYBRID, HYBRID is a SINGLETON class — which is taxonomically weak. Need at least one additional HYBRID observable to support 3-class partition over 2-class + HYBRID-singleton.

Honest framing: TRIPLE currently has 2 well-populated classes (Mersenne with $m_e + m_\tau$; Bergman with m_μ) + 1 singleton (HYBRID = Higgs alone). Three-class partition with one singleton is taxonomically a 2+1 not a 3-class.

REQUIRED FROM LYRA: identify at least one additional HYBRID-class observable beyond Higgs, OR honestly reframe TRIPLE as “Mersenne-edge / Bergman-bulk / HYBRID-singleton (Higgs only)”.

6. B1-B4 — Believability cross-checks

6.1. B1 — Precision believability

m_μ/m_e at 0.112%: confirmed substrate-natural form $207 = 200 + 7 = 2 \cdot 100 + g$, or $207 = 9 \cdot 23$ (not BST-primary). Bergman + Pochhammer cascade derivation believable structurally but PRE-STAGE only — Bergman cascade FORWARD content NOT YET filed at proper depth.

m_τ/m_μ at 0.06%: confirmed observed ratio 16.817 vs substrate via cross-family. Believable per cumulative Vol 2 work.

$m_H \approx 125.28$ GeV at 0.14%: confirmed observed 125.10 GeV (PDG 2022); 0.14% within Tier 2 STRUCTURAL range. Forward derivation FRAMEWORK-COMPLETE per Lyra Friday but not yet RIGOROUS.

6.2. B2 — Substrate-natural form believability

Class	Substrate primitive	BST-primary count
Mersenne	$M(p) = 2^p - 1$	direct substrate-tower
Bergman	$B(z, w)$ Bergman kernel	direct substrate-geometric
HYBRID	central vacuum + Higgs	substrate-K-type-spectral

All three classes use substrate-natural primitives (NOT ad hoc). B2 PASSES.

6.3. B3 — Multi-observable cross-coherence

Mersenne family observables ($m_e, m_\tau + \text{Higgs?}$): all consistent with Mersenne tower at $2^p - 1$ structure
Bergman family observables ($m_\mu + \text{cross-family } m_\tau/m_\mu$): all consistent with Bergman cascade

Concern: only 4 observables total are class-assigned in Lyra Friday ($m_\mu/m_e, m_\tau/m_\mu, m_H, EW$ 6). For a 3-class partition claim, 4-7 observables is sparse. Larger N would strengthen.

B3 PARTIAL — partition is supported but not over-determined yet.

6.4. B4 — Cross-mechanism-class coherence with substrate-Schur

K228+K225 substrate-Schur ($28 = 2 \cdot g\text{-rank}$) operates CROSS-CLASS (EW + inflation). If TRIPLE is correct, substrate-Schur is meta-primitive ABOVE TRIPLE.

If TRIPLE is incorrect (single-primitive underlies all), substrate-Schur is THE single primitive.

K229 disposition cross-coupled with K228+K225 substrate-Schur status. Both rise or fall together.

B4 PASSES coherence check but TIGHTLY COUPLED to K228+K225 disposition.

7. Candidate SEVENTH audit category — SUBSTRATE-MECHANISM-FAMILY-PARTITION

K226+K227+K228 added three observable-level categories (4-5-6 in expanded framework). K229 is candidate META-LEVEL category:

SUBSTRATE-MECHANISM-FAMILY-PARTITION (CANDIDATE 7th category): substrate IDENTIFIES a cross-observable taxonomy in which N observables sort into K mechanism classes ($K \geq 2$) by substrate-region or substrate-primitive use-pattern. Categories are meta-level (above individual observable predictions). Promotion to STANDING requires (i) substrate-FORCING argument for K, (ii) class-mutual-exclusivity verification, (iii) multi-observable coverage per class (no singleton classes), (iv) Cal cold-read closure.

K229 PRE-STAGE disposition for category: CANDIDATE category. Promotion to STANDING is DEFERRED until three gates closed (Section 0).

8. Disposition

8.1. Verdict

K229 PRE-STAGE — CONDITIONAL PASS at PATTERN-DISCOVERY tier; FRAMEWORK promotion DEFERRED pending three gates.

Conditional PASS rationale: - Forward derivations at Tier 2 STRUCTURAL precision are substantively present (m_μ/m_e 0.112%, m_τ/m_μ 0.06%, m_H 0.14%) - Substrate primitives invoked (Mersenne tower, Bergman

kernel, central K-type vacuum) are substrate-natural and NOT ad hoc - Cross-K-audit substrate-Schur link (K228+K225) substantively reinforces meta-classification at FRAMEWORK level - Honest framing operational: TRIPLE is currently 2-class + 1-singleton, not over-determined 3-class

Deferred items: - Class-definition gate (gen-indexing + edge/bulk K-type + HYBRID criterion) - Independence-vs-overlap gate (cross-family ratios + substrate-Schur meta-primitive scope) - Forcing-vs-pattern-discovery gate (substrate D_IV⁵ structural argument for K = 3)

8.2. Severity ratings on gaps

- Gate 1 class-definition: MODERATE severity. Pre-stage proceeds with tentative reading but Lyra clarification required before STANDING promotion
- Gate 2 independence-vs-overlap: MODERATE severity. Cross-family contamination is acknowledged honestly; HYBRID singleton is acknowledged honestly
- Gate 3 forcing-vs-pattern-discovery: CRITICAL severity for FRAMEWORK promotion. Without substrate-FORCING argument, TRIPLE remains CANDIDATE-tier descriptive

8.3. Comparison to Friday six-category framework

K229 is META-LEVEL (taxonomy across observables) where K222-K228 are OBSERVABLE-LEVEL (one prediction at a time). Promotion of K229 to STANDING expands the audit framework from observable-level (6 categories) to dual-level (6 observable + 1 meta).

This is a substantively new audit-framework expansion. Cal cold-read on the meta-category itself is warranted before STANDING promotion.

9. Cross-K-audit links

- K221 Lyra Strong-Uniqueness F22 v1.7 → v1.8 absorbing Cal #247 (parallel to K229: both architecture-tier)
- K225 Elie Toy 3861 substrate $n_s = 27/28$ PASS at EXACT-BOUND-SATISFYING — substrate-Schur composite $28 = 2 \cdot g\text{-rank}$ shared with K228; meta-primitive above TRIPLE
- K228 Elie Toy 3876 substrate $\alpha^{-1} = N_{\max} + 1/28$ PASS at STRUCTURAL-CORRECTION-TERM-IDENTIFICATION — meta-primitive above TRIPLE
- K229b $N_{c^2} \cdot n_C = 45$ parallel substrate primitive PMNS + CKM (Cal #36 Schur-generator, candidate filing separately)

Cross-K-audit substantive picture: K225 + K228 + K229b establish a SUBSTRATE-SCHUR LAYER (meta-primitives $28 = 2 \cdot g\text{-rank} + 45 = N_{c^2} \cdot n_C$) that operates ABOVE the K229 TRIPLE mechanism-class layer. The TRIPLE provides class-internal forward derivations; substrate-Schur provides cross-class correction terms.

This is a TWO-LAYER picture: substrate-Schur (meta-primitives) + TRIPLE (mechanism-class partition).

If validated at STANDING tier across both layers, this is the most substantively structured audit-framework finding of the Phase 1 K-audit cascade.

10. Actions

1. Lyra (Saturday-Sunday): respond to three class-definition clarifications (gen-indexing, edge/bulk K-type, HYBRID criterion). File F25 v0.1 substrate-mechanism class TRIPLE forcing argument if attempting FRAMEWORK promotion.
2. Keeper (Saturday): K229b candidate pre-stage filing ($N_{c^2} \cdot n_C = 45$ parallel primitive). K229 full audit when Lyra closes Gate 1 class-definition.
3. Grace (Saturday): G15 v0.6 Cluster L ($28 = 2 \cdot g\text{-rank}$) + G14 v0.3 $N_{c^2} \cdot n_C = 45$ addition; consider Cluster M candidate if TRIPLE-internal substrate-Schur generators distinct
4. Cal (Saturday-following): cold-read on K229 PRE-STAGE + meta-category candidate; Methodology Index v0.16 absorption of meta-level audit category

5. Elie (Saturday): HYBRID-class second-observable search (avoid singleton HYBRID); Vol 16 Ch 9 numerical support for Lyra primary

11. Substantive honest framing

K229 is a substrate-architectural CLAIM, not yet a substrate-architectural THEOREM. The Friday six-category audit framework was developed at observable-level; adding a meta-level category is a substantive framework expansion.

Pre-stage CONDITIONAL PASS reflects: - TRIPLE has genuine substantive content (concrete forward derivations + substrate-natural primitives + cross-K-audit cross-link) - TRIPLE is not yet substrate-FORCED at FRAMEWORK level (Gate 3 open) - TRIPLE may be a TWO-LAYER picture with substrate-Schur meta-primitives (K225 + K228 + K229b) ABOVE the mechanism-class partition

This is a substantive finding worth carrying forward at PRE-STAGE; promotion is multi-week Lyra + Keeper joint work.

Keeper K229 PRE-STAGE filing — Saturday 2026-06-06 10:15 EDT (**date-verified**). Conditional PASS at PATTERN-DISCOVERY tier; FRAMEWORK promotion deferred pending three gates. Two-layer substrate picture (substrate-Schur meta-primitives + TRIPLE mechanism-class partition) is the substantive synthesis. Ready for Lyra response on three class-definition clarifications.